

II CART classification for subscription churn

"Churn" means when a customer unsubscribes or cancels a contract.

⇒ A BI team wants to predict whether a customer will churn from a subscription platform.

The current root node contains 10 customers:

6 churn

4 no churn.

A candidate split by the binary feature

Support Tickets High = { yes, no }

produces:

t_L : 5 customers = 4 churn, 1 no churn

t_R : 5 customers = 2 churn, 3 no churn.

* Dataset

Customer	Support Tickets High	Churn	Destination Node
C01	YES	Churn	t_L
C02	YES	Churn	t_L
C03	YES	Churn	t_L
C04	YES	Churn	t_L
C05	YES	No Churn	t_L
C06	NO	Churn	t_R
C07	NO	Churn	t_R
C08	NO	No Churn	t_R
C09	NO	No Churn	t_R
C10	NO	No Churn	t_R

a) Compute the Gini impurity of the root node.

The initial root node contains 10 customers: 6 churn, 4 No.

The baseline proportions are:

$$p_{\text{churn}}(t) = \frac{6}{10} = 0.6$$

$$p_{\text{no churn}} = \frac{4}{10} = 0.4$$

$\Rightarrow$ Root node Gini impurity:

$$G(\text{root}) = 1 - \sum_{k=1}^K p_k(t)^2$$

$$= 1 - (0.6^2 + 0.4^2) = 1 - 0.52 = 0.480$$

b) Compute the Gini Impurity of t_L and t_R

For the right child node t_R , the proportions are:

$$p_{\text{churn}} = \frac{2}{5} = 0.4$$

$$p_{\text{no churn}} = \frac{3}{5} = 0.6$$

$$G(t_R) = 1 - \sum_{k=1}^K p_k(t_R)^2 = 1 - (0.4^2 + 0.6^2)$$

$$= 1 - (0.16 + 0.36) = 0.480$$

For the left child node t_L , the proportions are:

$$p_{\text{churn}} = \frac{4}{5} = 0.8$$

$$p_{\text{no churn}} = \frac{1}{5} = 0.2$$

$$G(t_L) = 1 - (0.8^2 + 0.2^2) = 1 - (0.64 + 0.04)$$

$$= 0.320$$

c) Compute the weighted post-split Gini impurity.

The weighted post-split impurity G_{split} accounts for the size variations in the resulting subjects by scaling each child node's impurity by its respective sample fraction relative to the total parent population

$$n = 10 \quad n_L = 5 \quad n_R = 5$$

$$G_{\text{split}} = \frac{n_L}{n} G(t_L) + \frac{n_R}{n} G(t_R)$$

$\Rightarrow$ Substituting:

$$G_{\text{split}} = \frac{5}{10} (0.320) + \frac{5}{10} (0.480)$$

$$= \frac{1}{2} (0.320) + \frac{1}{2} (0.480)$$

$$G_{\text{split}} = 0.160 + 0.240 = 0.400$$

d) Compute the impurity reduction and explain whether the split is useful.

$\Rightarrow$ The impurity reduction ΔG measures the structural improvement or gain in purity achieved by partitioning the root population.

$$\Delta G = G(\text{root}) - G_{\text{split}}$$

$$\Delta G = 0.480 - 0.400 = \underline{0.080}$$

$\Rightarrow$ It is useful. We have a positive impurity reduction.

- The left child node t_L successfully isolates a highly concentrated risk segment where 80% of the customers churn.
- The right child node t_R captures a retention segment where 60% of customers stay.

13] A regression tree predicts the number of monthly app logins from customer segment behavior. The target values at a parent node are given by a vector:

$$y = (5, 6, 7, 20, 22, 24, 25)$$

Two candidate splits are tested:

$$\text{Split A: } t_L: y = (5, 6, 7); \quad t_R: y = (20, 22, 24, 25)$$

$$\text{Split B: } t_L: y = (5, 6, 7, 20); \quad t_R: y = (22, 24, 25)$$

a) Compute the parent node mean and SSE.

The total sample size of the parent node is $n = 7$.

$$\begin{aligned}\bar{y}_t &= \frac{1}{7} \sum_{i \in t} y_i = \frac{5 + 6 + 7 + 20 + 22 + 24 + 25}{7} \\ &= \frac{109}{7} \approx 15.571\end{aligned}$$

$$SSE(t) = \sum_{i \in t} (y_i - \bar{y}_t)^2$$

$$\begin{aligned}SSE(t) &= (5 - 15.571)^2 + (6 - 15.571)^2 + (7 - 15.571)^2 + (20 - 15.571)^2 \\ &\quad + (22 - 15.571)^2 + (24 - 15.571)^2 + (25 - 15.571)^2\end{aligned}$$

$$SSE(t) = 497.714$$

b) Compute the SSE_{split} for split A.

Split A partitions the population into:

$$t_L: y = (5, 6, 7)$$

$$t_R: y = (20, 22, 24, 25)$$

1. Left child node t_L

$$\Rightarrow n_L = 3$$

$$\bar{y}_{t_L} = \frac{5 + 6 + 7}{3} = \frac{18}{3} = 6.000$$

$$SSE(t_L) = (5-6)^2 + (6-6)^2 + (7-6)^2 = 2.000$$

2. Right child node t_R

$$\Rightarrow n_R = 4 \quad \bar{y}_{t_R} = \frac{20 + 22 + 24 + 25}{4} = \frac{91}{4} = 22.750$$

$$SSE(t_R) = (20-22.75)^2 + (22-22.75)^2 + (24-22.75)^2 + (25-22.75)^2 = 14.750$$

3. Total Post-split SSE

$$\begin{aligned} A \quad SSE_{split} &= SSE(t_L) + SSE(t_R) \\ &= 2.000 + 14.750 = 16.750 \end{aligned}$$

c) Compute SSE_{split} for split B.

$\Rightarrow$ Split B partitions the population into
 $t_L: y = (5, 6, 7, 20)$; $t_R: y = (22, 24, 25)$

1. Left child node (t_L)

$$\Rightarrow n_L = 4 \quad \bar{y}_{t_L} = \frac{5 + 6 + 7 + 20}{4} = \frac{38}{4} = 9.500$$

$$SSE(t_L) = (5-9.5)^2 + (6-9.5)^2 + (7-9.5)^2 + (20-9.5)^2 = 149.000$$

2. Right child node (t_R)

$$\Rightarrow n_R = 3 \quad \bar{y}_{t_R} = \frac{22 + 24 + 25}{3} = \frac{71}{3} \approx 23.667$$

$$SSE(t_R) = (22-23.667)^2 + (24-23.667)^2 + (25-23.667)^2 = 4.667$$

$$\begin{aligned} 3. \quad SSE_{split} &= SSE(t_L) + SSE(t_R) \\ &= 149.000 + 4.667 \end{aligned}$$

$$B \quad SSE_{split} = 153.667$$

d) Which split does CART select?

The CART algorithm selects A

Reasoning: Split A yields a significantly larger reduction in SSE compared to split B

$\Rightarrow$ Business-wise, split A correctly distinguishes the lower-frequency user segment $y \leq 7$, from higher-frequency users ($y \geq 20$).

17

Cost-Complexity Pruning

A CART classification tree has the following candidate subtrees:

Tree	Training misclassification $R(T)$	Number of Leaves $ T $
T_1	10	2
T_2	7	4
T_3	5	8

$\Rightarrow$ Use the cost-complexity criterion

$$R_\alpha(T) = R(T) + \alpha |T|$$

to evaluate the subtrees.

a) Compute $R_\alpha(T_1)$, $R_\alpha(T_2)$, $R_\alpha(T_3)$ for $\alpha = 0$.

When $\alpha = 0$ there is no penalty associated with the complexity (size) of the tree.

$$T_1: R_0(T_1) = 10 + 0 = 10$$

$$T_2: R_0(T_2) = 7 + 0 = 7$$

$$T_3: R_0(T_3) = 5 + 0 = 5$$

Conclusion: Under a zero-penalty regime ($\alpha = 0$), the largest tree T_3 minimizes the total cost criterion because it achieves the lowest training error.

b) Compute the same values for $\alpha = 1$.

$$T_1: R_1(T_1) = 10 + 1(2) = 12$$

$$T_2: R_1(T_2) = 7 + 1(4) = 11$$

$$T_3: R_1(T_3) = 5 + 1(8) = 13$$

Conclusion: Under a complexity penalty of $\alpha = 1$, the intermediate tree T_2 minimizes the objective criterion ($R_1(T_2) = 11$). It achieves an optimal "Trade-off" by reducing training misclassification without introducing the high complexity penalty cost seen in T_3 .

c) Compute the same values for $\alpha = 2$.

$\Rightarrow$ Homework!

d) For each value of α state which tree is selected

$\Rightarrow$ homework!

e) Explain in BI language why increasing α favors simpler trees.

$\Rightarrow$ In BI and data analytics, the parameter α acts as a financial or strategic penalty weight on model complexity. Increasing α shifts the focus away from raw historical data and heavily favors smaller, simpler tree structures.

⚠ IMPORTANT: Larger trees tend to over-fit more than smaller ones!

$\Rightarrow$ Trade-off must be found!

19 Model Evaluation Metrics for Risk Detection

A company compares two CART models for detecting risky invoices. The test set results from their respective confusion matrices:

Model A: $TP = 20$, $FP = 10$, $TN = 70$, $FN = 5$

Model B: $TP = 17$, $FP = 4$, $TN = 76$, $FN = 8$

a) Compute the accuracy for both models.

⇒ The classification accuracy measures the proportion of total correct predictions relative to the entire population size.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

Model A:

$$A_A = \frac{20 + 70}{20 + 10 + 70 + 5} = \frac{90}{105} \approx 0.857 \text{ (85.7\%)}$$

Model B:

$$A_B = \frac{17 + 76}{17 + 4 + 76 + 8} = \frac{93}{105} \approx 0.886 \text{ (88.6\%)}$$

b) Calculate the precision for both models.

⇒ The precision (positive predictive value) measures the proportion of correctly identified risky invoices out of all invoices flagged as risky by the model:

$$\text{Precision} = \frac{TP}{TP + FP}$$

Model A: $P_A = \frac{20}{20 + 10} = \frac{20}{30} \approx 0.667 \text{ (66.7\%)}$

Model B: $P_B = \frac{17}{17 + 4} = \frac{17}{21} \approx 0.810 \text{ (81.0\%)}$

c) Compute Recall for both models.

⇒ Recall (sensitivity) measures the proportion of actual risky invoices that the model successfully flags.

$$\text{Recall} = \frac{TP}{TP + FN}$$

Model A: $R_A = \frac{20}{20 + 5} = \frac{20}{25} = 0.800 \text{ (80.0\%)}$

Model B: $R_B = \frac{17}{17 + 8} = \frac{17}{25} = 0.680 \text{ (68.0\%)}$

d) Compute the F_1 score for both models.

⇒ F_1 is a harmonic mean of precision and recall, balancing both metrics into a single metric.

* It is particularly useful when evaluating models on imbalanced datasets!

$$F_1 = 2 \left(\frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \right)$$

Model A: Using $P_A \approx 0.667$, $R_A = 0.800$

$$F_{1,A} = 2 \left(\frac{P_A R_A}{P_A + R_A} \right) = 2 \left(\frac{0.667 \cdot 0.800}{0.667 + 0.800} \right)$$

$$F_{1,A} \approx 0.727 \quad (72.7\%)$$

Model B: Using $P_B \approx 0.810$, $R_B = 0.680$

$$F_{1,B} = 2 \left(\frac{P_B R_B}{P_B + R_B} \right) = 2 \left(\frac{0.810 \cdot 0.680}{0.810 + 0.680} \right)$$

$$F_{1,B} \approx 0.739 \quad (73.9\%)$$

e) If a risky invoice is very expensive, which model should be preferred?

$\Rightarrow$ Model A. In an audit or fraud-detection context, letting a risky invoice slip through means a false negative (FN).

$\Rightarrow$ Choose model based on Recall !